

FACULTY ELECTRICAL ENGINEERING, MATHEMATICS
AND COMPUTER SCIENCE

Department of Microelectronics

[ET4371] Mixed-mode Wireless transceivers

Assignment 3

Bonus 1

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1 Bonus 1 resistive

What it is asked to be performed is the out-of-band matching technique. Specifically, by utilizing secondary mixing we can cancel the direct IM3 distortion. Basically, beside the direct generation of third order intermodulation products, the second order products and the baseband can mix with the fundamental products indirectly creating third order intermodulation product as well. Therefore, by controlling the voltage at the source we can control the indirect mixing and cancel the direct IM3 distortion, improving the linearity of the system.

A common technique is the use of source degeneration. I expect the inductive source degeneration to perform better than the resistive counterpart. In the course of the previous quarter we analyse a BJT transistor with source degeneration in terms of distortion; however, the same process could be performed for a FET. In this analysis we expressed the IM3 component with the help of Volterra series:

$$IM3 = \frac{3}{4} \frac{\hat{V}_S^2}{V_T^2} |H_1(s)|^3 \cdot |T(s)| \cdot \left| \left(\frac{1}{2} H_1(2s)T(2s) + H_1(\Delta s)T(\Delta s) - 1 \right) \right| \quad (1)$$

$$T(s) = 1 + sC_{JE}Z_e(s) \quad (2)$$

From equation 2 we can observe that choosing an inductive source degeneration can resonate out the capacitance and make the function $T(s)$ zero, reducing the IM3 component in a greater extent. Another important benefit of the inductive source degeneration is that it is a short for the DC, meaning that it doesn't consume voltage headroom, and it doesn't generate thermal noise like the resistance.

I present the results after the implementation of the inductive and resistive source degeneration in which it becomes apparent that the suppression has been improved. Moreover, the gain has been reduced since in a source degenerative stage the effective transconductance is smaller than the standard gain cell.

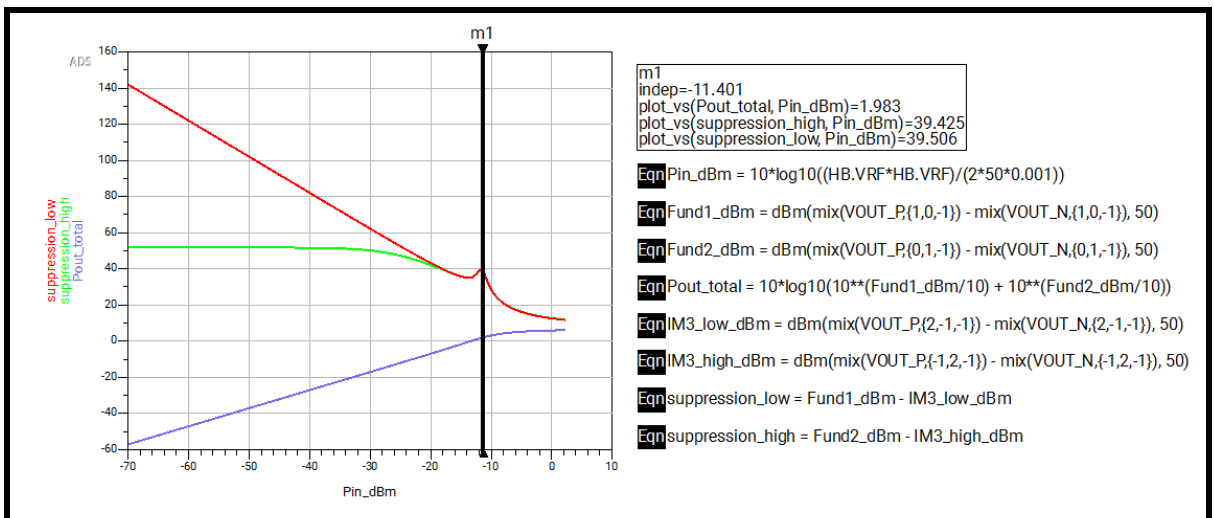


Figure 1: Resistive source degeneration.

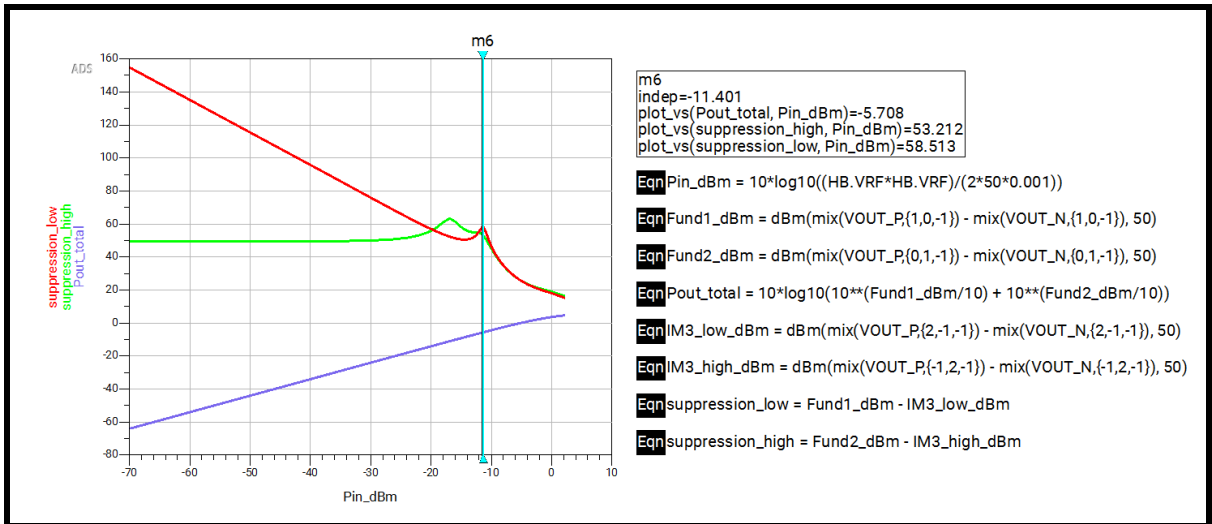


Figure 2: Inductive source degeneration.